

Lukáš Randuška

Institute of Experimental Physics, Slovak Academy of Sciences

Košice, Slovakia

lukasran28@gmail.com

26 June 2026

Dear ESWW2026 Selection Committee,

I am a Bachelor student of Data Science and Artificial Intelligence at Pavol Jozef Šafárik University in Košice. I am currently working part-time at the Institute of Experimental Physics, Slovak Academy of Sciences, in Košice.

For the past three years I have been researching coronal mass ejections (CMEs), the solar wind, and their interactions with one another and with the Earth — how they affect our planet and how their behaviour can be predicted. I am currently developing a CME detection-to-impact pipeline. It begins with a CACTus-inspired CME detection algorithm applied to LASCO C2 coronagraph images, reaching effectiveness on par with the original CACTus algorithm. My approach then ties this detection directly to a network that predicts the transit time of the CME from parameters observable at onset. This network is a physics-informed neural network, pre-trained on synthetic events generated as solutions of the extended drag-based model (EDBM). The detected velocity, onset time, and predicted transit time are then used to reconstruct a 1D time-series of the CME's transit velocity up to its arrival at Earth, which we expect will yield strong results in predicting the Dst index tens of hours into the future.

I am coming to ESWW2026 to present my work and to gain experience in communicating my research. The conference would let me present my results to the international space-weather community, receive constructive criticism from experts in the field, and find potential collaborators working on CME detection and forecasting. As an early-career researcher standing at the start of my path, I am also keen to see the current frontier of machine-learning research in space weather, which would directly help guide and inspire the direction of my own work and my plans for graduate study. This would be my first international conference, and attending it would be a formative step in my training.

Thank you for considering my application. I would be very grateful for the opportunity to take part in ESWW2026 and to contribute to the space-weather community.

Sincerely,

Lukáš Randuška